
PHYSICS FORMULAS

Master Formula Sheet | Chapters 1–14

Chapters Covered

1. Electric Charges and Fields
2. Electrostatic Potential and Capacitance
3. Electric Resistance and Ohm's Law
4. Moving Charges and Magnetism
5. Magnetism and Matter
6. Electromagnetic Induction
7. Alternating Current
8. Electromagnetic Waves
9. Ray Optics and Optical Instruments
10. Wave Optics
11. Dual Nature of Radiation and Matter
12. Atoms
13. Nuclei
14. Semiconductor Electronics

ISC Class XII Physics

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CHAPTER 1

Electric Charges and Fields

Formula Sheet + Tips & Tricks

Chapter 1: Electric Fields and Charges

Formula Sheet + Tips & Tricks

Formula Sheet

I. Electric Charge and Quantisation

Quantisation of charge

$$Q = ne$$

n = number of electrons, $e = 1.6 \times 10^{-19} \text{ C}$

II. Coulomb's Law and Superposition

Coulomb's Law

$$F = k \frac{q_1 q_2}{r^2}$$

$k = 9 \times 10^9 \text{ N m}^2\text{C}^{-2}$; like charges repel, unlike attract

Force due to g

$$F = k \frac{e^2}{r^2} = mg$$

used to find separation at which electrostatic force equals weight

Superposition principle

$$\vec{F}_{net} = \sum \vec{F}_i$$

vector sum of individual Coulomb forces on a charge

III. Equilibrium of Charges

Null point condition

$$k \frac{q_1 q_3}{r_1^2} = k \frac{q_2 q_3}{r_2^2}$$

net force on q_3 is zero; null point lies on the side of the smaller charge if signs are opposite, and between the charges if signs are the same

IV. Electric Field Lines and Intensity

Electric field (point charge)

$$E = k \frac{q}{r^2}$$

field points away from $+q$, towards $-q$

Force on a charge in a field

$$F = qE$$

direction along E for $+q$, opposite for $-q$

Field in vector form

$$\vec{E} = k \frac{q}{r^3} \vec{r}$$

$\vec{r}$ = position vector from source charge to field point

V. Electric Dipole

Dipole moment

$$p = q \times 2l$$

$2l$ = separation between the two charges

Axial (end-on) field

$$E_{\parallel} = \frac{2kp}{r^3}$$

valid for $r \gg l$

Equatorial (broadside-on) field

$$E_{\perp} = \frac{kp}{r^3}$$

valid for $r \gg l$

Torque on a dipole

$$\tau = pE \sin \theta$$

maximum at $\theta = 90^\circ$: $\tau_{max} = pE$

VI. Electric Flux and Gauss's Law

Electric flux

$$\phi = \vec{E} \cdot \vec{A} = EA \cos \theta$$

θ = angle between $\vec{E}$ and the outward normal $\hat{n}$

Gauss's Law

$$\phi = \frac{q_{enc}}{\epsilon_0}$$

flux depends only on charge enclosed, not on the size/shape of the surface

VII. Linear Charge Distribution

Field due to infinite line charge

$$E = \frac{\lambda}{2\pi\epsilon_0 r} = \frac{2k\lambda}{r}$$

λ = linear charge density (C/m)

VIII. Sheet(s) of Charge

Field due to a charged sheet

$$E = \frac{\sigma}{2\epsilon_0}$$

field due to one plate; σ = surface charge density

Field between oppositely charged plates

$$E = \frac{\sigma}{\epsilon_0}$$

fields of both plates add between them and cancel outside

IX. Charged Spherical Shell

Charge from surface density

$$Q = 4\pi R^2 \sigma$$

R = radius of sphere

Field of a charged conducting sphere

$$E_{in} = 0; \quad E_{surface} = k \frac{Q}{R^2}; \quad E_{out} = k \frac{Q}{r^2}$$

all charge behaves as if concentrated at the centre for points outside

Tips, Tricks & Hacks

- $k = \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2\text{C}^{-2}$ — memorise this once, it drives every Coulomb's law and field question in the chapter.
- If a question gives “force is doubled” or “distance is halved”, use ratios directly: $F \propto \frac{1}{r^2}$, don't recompute from scratch.
- For a charge in equilibrium under gravity + electric force (classic “bead on thread” or “charged ball” problems), always resolve forces along and perpendicular to the string/incline before equating.
- Dipole field: remember End-on = $2 \times$ Broadside-on at the same distance ($E_{\parallel} = 2E_{\perp}$ for $r \gg l$) — a very common one-line MCQ trick.
- Gauss's law $\phi = \frac{q_{enc}}{\epsilon_0}$ only cares about *enclosed* charge — charges outside the surface contribute zero net flux, even though they still create field at points on the surface.
- For infinite sheet/cylinder/sphere symmetry problems, choose a Gaussian surface that matches the symmetry (pillbox for sheet, cylinder for wire, sphere for point/spherical charge) — this is 90% of the trick.
- Field inside a conductor is always zero in electrostatics; all excess charge resides on the outer surface only.
- Quick sanity check: field lines start on + charges, end on − charges, never cross, and are denser where the field is stronger.

CHAPTER **2**

Electrostatic Potential and Capacitance

Formula Sheet + Tips & Tricks

Chapter 2: Electrostatic Potential and Capacitance

Formula Sheet + Tips & Tricks

Formula Sheet

I. Electric Potential and Potential Difference

Potential due to a point charge

$$V = \frac{kQ}{r}$$

$V \propto 1/r$; slope of a V vs $1/r$ graph gives kQ

Potential difference & work

$$V = \frac{W}{q}; \quad W = q(V_B - V_A)$$

W = work done in moving charge q between two points

Potential due to several charges

$$V = \sum_i \frac{kQ_i}{r_i}$$

scalar sum (superposition), unlike vector sum for $\vec{E}$

Kinetic energy gained

$$K = qV$$

gain in KE = work done by the field for a charge accelerated through PD V

II. Potential Gradient

Potential gradient

$$E = \frac{\Delta V}{d}$$

electric field = (negative of) rate of change of potential with distance; potential decreases along $\vec{E}$

III. Electrostatic Potential Energy

PE of two point charges

$$U = \frac{kq_1q_2}{r}$$

energy needed to assemble the charge pair from infinity

Energy in electron-volt

$$1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$$

convenient energy unit at atomic scale

IV. Electric Dipole (Energy Aspects)

Dipole moment

$$p = q \times 2l$$

$2l$ = separation between the two charges of the dipole

Torque on a dipole

$$\tau = pE \sin \theta$$

maximum at $\theta = 90^\circ$: $\tau_{max} = pE$

PE of a dipole in a field

$$U = -pE \cos \theta$$

$U = 0$ at $\theta = 90^\circ$; minimum ($-pE$) at $\theta = 0^\circ$

Work done rotating a dipole

$$W = pE(\cos \theta_1 - \cos \theta_2)$$

work to rotate dipole from θ_1 to θ_2 in the field

V. Capacitance of a Conductor

Capacitance

$$C = \frac{Q}{V}$$

ratio of charge to potential; unit: farad (F)

Capacitance of an isolated sphere

$$C = 4\pi\epsilon_0 R = \frac{R}{k}$$

R = radius of the sphere

VI. Parallel Plate Capacitor

Parallel plate capacitance (air)

$$C = \frac{\epsilon_0 A}{d}$$

A = plate area, d = plate separation

Charge on capacitor

$$Q = CV$$

V = potential difference applied across the plates

VII. Capacitor with Dielectric Slab

Capacitance with dielectric

$$C = \frac{K\epsilon_0 A}{d}$$

K = dielectric constant of the medium between the plates

Effect of dielectric on C , Q , V

Q = const if battery removed; V = const if battery connected
 battery removed $\Rightarrow V$ falls, Q fixed; battery connected $\Rightarrow V$ fixed, Q rises

VIII. Energy of a Charged Capacitor

Energy stored

$$U = \frac{1}{2}CV^2 = \frac{Q^2}{2C} = \frac{1}{2}QV$$

use whichever form matches the known quantities (Q fixed or V fixed)

Energy density

$$u = \frac{U}{\text{Volume}} = \frac{1}{2}\epsilon_0 E^2$$

energy stored per unit volume of the field region

IX. Combination of Capacitors

Series combination

$$\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$$

same charge Q on each capacitor; PDs add up

Parallel combination

$$C_p = C_1 + C_2 + \dots$$

same PD across each capacitor; charges add up

X. Miscellaneous**Potential of a charged sphere**

$$V_{\text{surface}} = \frac{kQ}{R}; \quad V_{\text{inside}} = V_{\text{surface}} \text{ (constant)}$$

for a hollow/solid conducting sphere, potential inside equals that on the surface

Balancing a charged drop

$$qE = mg$$

condition for a charged drop to remain stationary between plates

Tips, Tricks & Hacks

- $V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$ is a scalar — superpose potentials by simple algebraic addition (with sign), unlike fields which need vector addition.
- Potential is constant on an equipotential surface, and equipotentials are always $\perp$ to field lines — use this to sketch quickly.
- $E = -\frac{dV}{dr}$: if V vs r graph is given, the field is (minus) the slope — a favourite graph-based MCQ.
- For capacitors in series, charge Q is the same on each; in parallel, voltage V is the same on each. Memorise this pairing to avoid mixing up formulas.
- Energy stored $U = \frac{1}{2}CV^2 = \frac{1}{2}QV = \frac{Q^2}{2C}$ — pick the version that matches what's held constant (battery connected $\Rightarrow V$ fixed; battery disconnected $\Rightarrow Q$ fixed) before and after inserting a dielectric/changing plate separation.
- Dielectric slab fully filling the gap: $C' = KC_0$. Partially filling: treat as series combination of air gap and dielectric gap.
- Common trap: after charging and then *disconnecting* the battery, Q stays constant, not V ; if the battery stays connected, V stays constant, not Q . Re-derive energy change from whichever is fixed.
- Work done in moving a charge on the *same* equipotential surface is always zero, regardless of the path.

CHAPTER 3

Electric Resistance and Ohm's Law

Formula Sheet + Tips & Tricks

Chapter 3: Electric Resistance and Ohm's Law

Formula Sheet + Tips & Tricks

Formula Sheet

I. Electric Current and Current Density

Current and charge

$$Q = It$$

basic definition of steady current

Number of electrons

$$n = \frac{Q}{e}$$

$$e = 1.6 \times 10^{-19} \text{ C}$$

Variable current

$$Q = \int_{t_1}^{t_2} i \, dt$$

equals the area under the $I-t$ graph

II. Drift Velocity and Mobility

Microscopic form of current

$$I = nAev_d \Rightarrow v_d = \frac{I}{nAe}$$

n = free-electron density

III. Resistance and Resistivity

Ohm's Law

$$V = IR \Rightarrow R = \frac{V}{I}$$

defines resistance R

Resistance and resistivity

$$R = \rho \frac{L}{A} \Rightarrow \rho = \frac{RA}{L}$$

$$\text{conductivity } \sigma = \frac{1}{\rho}$$

Current density form

$$J = \sigma E = \frac{E}{\rho}$$

$$\rho = \frac{E}{J}$$

IV. Change in Resistance on Stretching the Wire

Stretching a wire

$$A_1 L_1 = A_2 L_2$$

if length becomes n times, new resistance $R' = n^2 R$

V. Effect of Temperature on Resistance and Resistivity

Temperature dependence of resistance

$$R_T = R_0(1 + \alpha T)$$

$$\alpha = \frac{R_T - R_0}{R_0 T} = \text{temperature coefficient of resistance}$$

VI. Combination of Resistances

Resistors in series

$$R_{\text{eq}} = R_1 + R_2 + \cdots + R_n$$

current is the same through each resistor

Resistors in parallel

$$\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_n}$$

voltage is the same across each resistor

VII. Electric Power and Energy

Electric power

$$P = VI = I^2 R = \frac{V^2}{R}$$

electrical energy $E = Pt$

VIII. Electric Cell

Terminal potential difference

$$V = \varepsilon - Ir \Rightarrow r = \frac{\varepsilon - V}{I}$$

while charging: $V = \varepsilon + Ir$

IX. Combination of Cells

Cells in series

$$\varepsilon_{\text{total}} = \sum \varepsilon_i, \quad r_{\text{total}} = \sum r_i$$

for n cells connected in series

Cells in parallel

$$\varepsilon_{\text{eff}} = \varepsilon, \quad r_{\text{eff}} = \frac{r}{n}$$

for n identical cells connected in parallel

X. Kirchhoff's Laws

Kirchhoff's Current Law

$$\sum I_{\text{in}} = \sum I_{\text{out}}$$

junction rule

Kirchhoff's Voltage Law

$$\sum \varepsilon = \sum IR$$

loop rule, around any closed loop

XI. Wheatstone Bridge and Metre Bridge

Balanced Wheatstone bridge

$$\frac{P}{Q} = \frac{R}{S}$$

no current flows through the galvanometer

Metre bridge

$$\frac{R}{S} = \frac{l}{100 - l}$$

l = balance length measured from the left end

XII. Potentiometer

Potential gradient

$$k = \frac{\varepsilon}{R_w + r} \times \frac{R_w}{L}$$

potential drop per unit length (V/m)

Potentiometer balance condition

$$\frac{\varepsilon_1}{\varepsilon_2} = \frac{l_1}{l_2}$$

emf is proportional to balance length

XIII. Miscellaneous

Circuit problem-solving

Ohm's Law + series-parallel reduction + Kirchhoff's Laws
apply in combination, as per the circuit topology

Tips, Tricks & Hacks

- Drift velocity is tiny ($\sim 10^{-4}$ m/s) — if an answer for v_d comes out large, you've likely mixed up n (electron density, $\sim 10^{28}$ m $^{-3}$) with something else.
- Colour-code the two big “series vs parallel” rules: series adds resistances and keeps current the same; parallel adds conductances (reciprocals) and keeps voltage the same.
- Wheatstone bridge is balanced ($\frac{P}{Q} = \frac{R}{S}$) $\Rightarrow$ galvanometer carries no current $\Rightarrow$ you can just delete that branch when redrawing the circuit.
- Metre bridge is a Wheatstone bridge in disguise: $\frac{R}{S} = \frac{l}{100 - l}$ — always measure l from the same end as R .
- In a potentiometer, the balancing length l is directly proportional to emf *only when no current is drawn* from the cell being measured — that's the whole point of using a potentiometer over a voltmeter.
- For power problems, choose the right form: if R is fixed and V varies, use $P = V^2/R$; if I is fixed, use $P = I^2 R$ — picking the wrong one wastes time squaring the wrong variable.
- Stretching a wire (volume constant): if length becomes $n\times$, resistance becomes $n^2\times$ (since $R \propto L^2$ when $V = AL$ is fixed) — a guaranteed 1-mark question.
- In cell combinations, series is best when internal resistance $\ll$ external resistance;

parallel is best when internal resistance $\gg$ external resistance — know why, it's a common conceptual question.

CHAPTER 4

Moving Charges and Magnetism

Formula Sheet + Tips & Tricks

Chapter 4: Moving Charges and Magnetism

Formula Sheet + Tips & Tricks

Formula Sheet

Constants Used Throughout This Chapter

$$\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1} \quad e = 1.6 \times 10^{-19} \text{ C}$$

$$m_e = 9.1 \times 10^{-31} \text{ kg} \quad m_p = 1.67 \times 10^{-27} \text{ kg}$$

$$m_\alpha = 6.7 \times 10^{-27} \text{ kg} \quad g = 9.8 \text{ m s}^{-2}$$

μ_0 = permeability of free space; e = elementary charge; m_e, m_p, m_α = rest masses of electron, proton, α -particle.

I. Magnetic Field due to Current (Biot–Savart & Ampere)

Field due to infinite straight wire

$$B = \frac{\mu_0 I}{2\pi r}$$

I = current, r = perpendicular distance from wire.

Field at centre of a circular coil

$$B = \frac{\mu_0 N I}{2r}$$

N = number of turns, r = radius of coil.

Field on the axis of a circular loop

$$B = \frac{\mu_0 I a^2}{2(a^2 + x^2)^{3/2}}$$

a = radius of loop, x = distance of point from centre along axis.

Field inside a long solenoid

$$B = \mu_0 n I$$

n = number of turns per unit length = N/L .

Field due to a moving point charge

$$B = \frac{\mu_0 q v \sin \theta}{4\pi r^2}$$

q = charge, v = speed, θ = angle between v and line joining charge to point.

II. Magnetic Force on a Moving Charge

Lorentz magnetic force

$$F = qvB \sin \theta$$

q = charge, v = speed, B = field, θ = angle between $\vec{v}$ and $\vec{B}$.

III. Path of a Charged Particle in a Uniform Field

Radius of circular path

$$r = \frac{mv}{qB}$$

m = mass, v = speed (perpendicular to B).

Time period & frequency of revolution

$$T = \frac{2\pi m}{qB}, \quad f = \frac{qB}{2\pi m}$$

Independent of the speed v of the particle.

Speed from accelerating potential

$$v = \sqrt{\frac{2qV}{m}}$$

V = accelerating potential difference.

Oblique entry (helical/spiral path)

$$r' = \frac{mv}{qB \sin \theta}$$

θ = angle between velocity and field; $\theta = 90^\circ$ gives circular path.

IV. Magnetic Force on a Current-Carrying Conductor

Force on a current-carrying conductor

$$F = BIL \sin \theta$$

L = length of conductor, θ = angle between current direction and B .

Microscopic (drift velocity) form

$$I = nAv_d e$$

n = free electron density, A = area of cross-section, v_d = drift velocity.

V. Force between Parallel Straight Current-Carrying Conductors

Force per unit length between two wires

$$\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi d}$$

d = separation between wires; like currents attract, unlike currents repel.

VI. Cyclotron**Cyclotron magnetic field**

$$B = \frac{2\pi m f}{q}$$

f = frequency of the applied oscillating voltage (= revolution frequency).

Maximum velocity & energy

$$v_{max} = 2\pi f R, \quad E_{max} = \frac{1}{2} m v_{max}^2$$

R = radius of the dees.

VII. Torque on a Current-Carrying Coil in a Magnetic Field**Torque on a current loop**

$$\tau = N B I A \sin \theta$$

A = area of coil, θ = angle between the normal to coil and $\vec{B}$.

Magnetic dipole moment

$$m = N I A$$

Torque may also be written $\tau = m B \sin \theta$; maximum torque $\tau_{max} = m B$ when $\theta = 90^\circ$.

VIII. Moving Coil Galvanometer: Sensitivities**Current sensitivity**

$$I_s = \frac{N B A}{k}$$

k = torsional constant of the suspension.

Voltage sensitivity

$$V_s = \frac{I_s}{R} = \frac{N B A}{k R}$$

R = resistance of the galvanometer coil.

IX. Conversion of Galvanometer into Ammeter/Voltmeter

Shunt resistance (galvanometer $\rightarrow$ ammeter)

$$S = \frac{I_g R_g}{I - I_g}$$

R_g = galvanometer resistance, I_g = full-scale deflection current, I = desired ammeter range.

Series resistance (galvanometer $\rightarrow$ voltmeter)

$$R = \frac{V}{I_g} - R_g$$

V = desired voltmeter range.

Net (effective) resistance

$$R_{\text{ammeter}} = \frac{R_g S}{R_g + S}, \quad R_{\text{voltmeter}} = R_g + R$$

Ammeter resistance is always very small; voltmeter resistance is always very large.

Tips, Tricks & Hacks

- Biot-Savart law gives $d\vec{B}$ due to a current element; always check the direction with the right-hand (screw) rule before plugging into the formula.
- For a straight wire, field $\propto \frac{1}{r}$ (not r^2) — don't confuse with the electric field/Coulomb pattern.
- For a circular loop, field at the centre is often asked to be compared with field on the axis ($B_{\text{axis}} = \frac{\mu_0 I R^2}{2(R^2 + x^2)^{3/2}}$) — setting $x = 0$ must reduce to the centre formula; use this to check your derivation.
- Force on a moving charge: $\vec{F} = q\vec{v} \times \vec{B}$ — force is always $\perp$ to velocity, so it changes direction, not speed $\Rightarrow$ path is a circle (or helix if there's a velocity component along $\vec{B}$).
- Radius of circular path $r = \frac{mv}{qB}$ and time period $T = \frac{2\pi m}{qB}$ — notice T is independent of speed, which is exactly why a cyclotron works at fixed frequency.
- Force between two parallel current-carrying wires: same direction currents attract, opposite directions repel — opposite to the “like charges repel” intuition from electrostatics, a classic trap.
- Moving coil galvanometer: current sensitivity = $\frac{NBA}{k}$, voltage sensitivity = $\frac{NBA}{kR}$ — increasing turns N boosts current sensitivity but also raises coil resistance, so

voltage sensitivity may not increase proportionally.

- To convert galvanometer to ammeter, connect a small shunt in *parallel*; to convert to voltmeter, connect a large resistance in *series* — remembering “ammeter = parallel shunt, voltmeter = series resistor” avoids 90% of mistakes here.

CHAPTER 5

Magnetism and Matter

Formula Sheet + Tips & Tricks

Chapter 5: Magnetism and Matter

Formula Sheet + Tips & Tricks

Formula Sheet

Physical Constants Used in This Chapter

Permeability of free space	$\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$
$\frac{\mu_0}{4\pi}$	$= 1 \times 10^{-7} \text{ T m A}^{-1}$
Conversion	$1 \text{ G} = 10^{-4} \text{ T}$

Section I: Magnetic Dipole, Magnetic Moment, etc.

Torque on a Magnetic Dipole

$$\tau = mB \sin \theta$$

m = magnetic moment, B = magnetic field, θ = angle between m and B .

Potential Energy of a Dipole

$$U = -mB \cos \theta = -\vec{m} \cdot \vec{B}$$

Minimum (most stable) when $\theta = 0^\circ$; maximum (unstable) when $\theta = 180^\circ$.

Axial Field of a Short Bar-Magnet

$$B_{\text{axis}} = \frac{\mu_0}{4\pi} \cdot \frac{2m}{r^3}$$

r = distance from centre of magnet along its axis.

Equatorial Field of a Short Bar-Magnet

$$B_{\text{eq}} = \frac{\mu_0}{4\pi} \cdot \frac{m}{r^3}$$

r = distance from centre of magnet along the equatorial line.

Magnetic Moment of an Equivalent Current Loop

$$m = NIA$$

N = number of turns, I = current, A = area of the loop.

Section II: Earth Magnetism

Components of Earth's Magnetic Field

$$B_E = \sqrt{H^2 + V^2}, \quad \tan \delta = \frac{V}{H}$$

H = horizontal component, V = vertical component, δ = angle of dip.

Magnetic Moment of a Uniformly Magnetised Rod

$$m = M \times \text{Volume} = M(\pi r^2 L)$$

M = intensity of magnetisation, r = radius, L = length of the rod.

Permeability and Susceptibility

$$\mu = \frac{B}{H}, \quad \mu_r = \frac{\mu}{\mu_0}, \quad \chi = \mu_r - 1$$

B = magnetic field induction, H = magnetising field, μ_r = relative permeability, χ = magnetic susceptibility.

Intensity of Magnetisation

$$M = \chi H$$

χ = magnetic susceptibility, H = magnetising field.

Curie's Law (Paramagnetic Materials)

$$\chi \propto \frac{1}{T}, \quad \frac{M}{B} \propto \frac{1}{T}$$

T = absolute temperature; susceptibility (and magnetic moment per unit field) rises as T falls.

Tips, Tricks & Hacks

- Bar magnet formulas mirror electric dipole formulas one-to-one: swap $p \leftrightarrow m$ (magnetic moment) and $\frac{1}{4\pi\epsilon_0} \leftrightarrow \frac{\mu_0}{4\pi}$ — if you know Chapter 1's dipole results, you already know most of this chapter.
- Axial field is twice the equatorial field at the same distance, exactly like the electric dipole case.
- Earth's magnetic field has three elements: horizontal component H , dip angle δ (angle of dip), and declination — $\tan \delta = \frac{V}{H}$ where V is the vertical component; memorising this one relation solves most numericals.
- At the magnetic poles, dip = 90° (field purely vertical); at the magnetic equator,

dip = 0° (field purely horizontal).

- Diamagnetic, paramagnetic, ferromagnetic — remember with “Dia repels/pushed out, Para & Ferro attract/pulled in”, and only ferromagnetics retain magnetism (hysteresis) after the field is removed.
- Curie’s law: paramagnetic susceptibility $\chi \propto \frac{1}{T}$ — as temperature rises, paramagnetism weakens (thermal agitation randomises the dipoles).
- Time period of oscillation of a magnet in a magnetic field: $T = 2\pi\sqrt{\frac{I}{mB}}$ — structurally identical to a torsion pendulum/SHM formula, so treat it as SHM.

CHAPTER 6

Electromagnetic Induction

Formula Sheet + Tips & Tricks

Chapter 6: Electromagnetic Induction

Formula Sheet + Tips & Tricks

Formula Sheet

I. Magnetic Flux and Faraday's Laws

Formula: $\Phi = BA \cos \theta$

Φ = magnetic flux, B = field strength, A = area, θ = angle between B and the normal to the coil.

Formula: $\varepsilon = -N \frac{d\Phi}{dt}$

ε = induced emf (Faraday's Law), N = number of turns, $d\Phi/dt$ = rate of change of flux per turn.

Formula: $q = \frac{N\Delta\Phi}{R}$

q = induced charge, $N\Delta\Phi$ = change in flux linkage, R = total circuit resistance.

Formula: $M = \mu_0 n N A, \quad \varepsilon = M \frac{dI}{dt}$

M = mutual inductance of a small coil (N turns, area A) wound on a long solenoid with n turns/metre.

II. Motional EMF

Formula: $\varepsilon = Bvl$

ε = motional emf, v = velocity of the conductor, l = length of conductor, valid when v , B , l are mutually perpendicular.

Formula: $t = \frac{d}{v}$

Duration for which emf lasts while a loop of side d (measured along the direction of motion) leaves a field region at speed v .

III. Self-Inductance

Formula: $\varepsilon = -L \frac{dI}{dt}$

ε = self-induced emf, L = coefficient of self-induction, dI/dt = rate of change of current.

Formula: $L = \frac{N\Phi}{I}$

L = self-inductance, $N\Phi$ = total flux linkage of the coil, I = current producing it.

Formula: $M = \frac{\mu_0 N_1 N_2 A}{l}$

Mutual inductance of two coils (N_1 , N_2 turns) wound on the same solenoid of length l and cross-section A .

IV. Mutual Induction

Formula: $M = \frac{\phi_1}{I_2}, \quad B_2 = \frac{\mu_0 I_2 R_2^2}{2(R_2^2 + x^2)^{3/2}}$

M = mutual inductance between two coaxial circular loops; R_2 = radius of larger loop, x = distance between centres, found via reciprocity by placing current I_2 in the larger loop and computing flux through the smaller loop.

Note: $\mu_0 = 4\pi \times 10^{-7} \text{ H m}^{-1}$ (also written $\text{Wb A}^{-1} \text{ m}^{-1}$ or T m A^{-1} or N A^{-2}).

Tips, Tricks & Hacks

- Lenz's law is just energy conservation in disguise: the induced current always opposes the *change* in flux, never the flux itself — state “opposes the change”, not “opposes the field”, to avoid losing marks.
- $\varepsilon = -\frac{d\phi}{dt}$: the minus sign is Lenz's law baked into the formula; for magnitude-only numericals you can drop it, but never in a conceptual answer.
- Motional emf $\varepsilon = Blv$ applies to a rod moving with velocity $\perp$ to both its length and the field — if the rod moves at an angle, use only the component of velocity $\perp$ to the field and rod.
- Self-inductance $L = \frac{N\phi}{I}$ behaves like “electrical inertia” — it opposes *change* in current, exactly analogous to mass opposing change in velocity.
- Energy stored in an inductor: $U = \frac{1}{2}LI^2$ — structurally identical to $\frac{1}{2}mv^2$ (kinetic energy analogy), useful for quick recall.
- Mutual inductance is symmetric: $M_{12} = M_{21}$, so it doesn't matter which coil you

call “primary” when calculating M .

- For a rod sliding on rails (classic EMI problem), always check whether it’s asked for emf, current, or the retarding force $F = BIl = \frac{B^2 l^2 v}{R}$ — these look similar but answer different questions.

CHAPTER 7

Alternating Current

Formula Sheet + Tips & Tricks

Chapter 7: Alternating Current

Formula Sheet + Tips & Tricks

Formula Sheet

I. Mean, Peak and rms Values of AC

rms and mean value (sinusoidal AC)

$$I_{rms} = \frac{I_0}{\sqrt{2}}, \quad V_{rms} = \frac{V_0}{\sqrt{2}}, \quad I_{mean} = \frac{2I_0}{\pi}$$

I_0, V_0 = peak current/voltage; mean value is defined only over a half-cycle (it is zero over a full cycle).

rms value of a sine + cosine combination

$$I = A \sin \omega t + B \cos \omega t \Rightarrow I_0 = \sqrt{A^2 + B^2}, \quad I_{rms} = \frac{I_0}{\sqrt{2}}$$

A, B = amplitudes of the sine and cosine components (same frequency).

rms value with a DC offset

$$I = I_{dc} + I_{ac} \sin \omega t \Rightarrow I_{rms} = \sqrt{I_{dc}^2 + \frac{I_{ac}^2}{2}}$$

I_{dc} = steady (DC) part, I_{ac} = peak value of the varying part.

II. AC Circuit with either of R, L or C

Inductive and capacitive reactance

$$X_L = \omega L = 2\pi f L, \quad X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$$

X_L = inductive reactance, X_C = capacitive reactance, f = frequency, L = inductance, C = capacitance.

Phase relations for pure elements

Pure R : V, I in phase. Pure L :

V leads I by 90° . Pure C : I leads V by 90°

Governs the phase relation between voltage and current for a single circuit element.

III. AC Circuit containing R and L

Impedance and phase angle (R-L series)

$$Z = \sqrt{R^2 + X_L^2}, \quad \tan \phi = \frac{X_L}{R}, \quad I = \frac{V}{Z}$$

Z = impedance, ϕ = phase angle between V and I (current lags voltage).

IV. AC Circuit containing R and C

Impedance and phase angle (R-C series)

$$Z = \sqrt{R^2 + X_C^2}, \quad \tan \phi = \frac{X_C}{R}$$

Same impedance triangle as R-L, but with I leading V instead of lagging.

V. L-C-R Circuit

Impedance and phase angle (series L-C-R)

$$Z = \sqrt{R^2 + (X_L - X_C)^2}, \quad \tan \phi = \frac{X_L - X_C}{R}$$

If $X_L > X_C$ the circuit is inductive (current lags); if $X_C > X_L$ it is capacitive (current leads).

Resonance condition

$$X_L = X_C \Rightarrow f_0 = \frac{1}{2\pi\sqrt{LC}}, \quad Z_{min} = R$$

f_0 = resonant frequency; at resonance the circuit is purely resistive and current is maximum.

Quality factor

$$Q = \frac{\omega_0 L}{R} = \frac{1}{R} \sqrt{\frac{L}{C}}$$

Q = quality factor, a measure of the sharpness of resonance.

VI. L-C Circuits and Oscillations

Series L-C circuit

$$V = |V_L - V_C|, \quad Z = |X_L - X_C|$$

In a pure L-C series circuit, V_L and V_C are 180° out of phase, so they subtract, not add.

Minimum-impedance / maximum-current condition

$$\omega L = \frac{1}{\omega C}$$

Used to find the L or C needed to minimise impedance (maximum brightness/current).

VII. Power Consumed in AC Circuits

Average power and power factor

$$P = V_{rms} I_{rms} \cos \phi, \quad \cos \phi = \frac{R}{Z}$$

For a pure resistor $\phi = 0$, so $P = I_{rms}^2 R = V_{rms} I_{rms}$.

VIII. Transformer

Ideal transformer relation

$$\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}, \quad \eta = \frac{P_{out}}{P_{in}} \times 100\%$$

N_p, N_s = primary/secondary turns; V_p, V_s, I_p, I_s = primary/secondary voltage and current; η = efficiency.

Tips, Tricks & Hacks

- rms value is what all AC meters read and what appears in power formulas: $I_{rms} = \frac{I_0}{\sqrt{2}}$, $V_{rms} = \frac{V_0}{\sqrt{2}}$ — never plug peak values directly into $P = VI$.
- Phase memory trick “CIVIL”: in a Capacitor, I leads V; in an Inductor, V leads I — instantly tells you the sign of the phase angle without re-deriving.
- Reactance formulas: $X_L = \omega L$ (grows with frequency), $X_C = \frac{1}{\omega C}$ (shrinks with frequency) — at very high frequency an inductor blocks and a capacitor passes; at $\omega \rightarrow 0$ it's the reverse.
- Resonance in LCR circuit occurs when $X_L = X_C$, i.e. $\omega_0 = \frac{1}{\sqrt{LC}}$ — at resonance impedance is purely resistive ($Z = R$) and current is maximum.
- Power factor $\cos \phi = \frac{R}{Z}$: a purely resistive circuit has $\cos \phi = 1$ (max power dissipated); a purely inductive or capacitive circuit has $\cos \phi = 0$ (zero average power, the “wattless current”).
- Quality factor $Q = \frac{\omega_0 L}{R} = \frac{1}{R} \sqrt{\frac{L}{C}}$ tells you how sharp the resonance is — higher

Q means a narrower resonance peak.

- Transformer: $\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$ — note current ratio is *inverted* compared to voltage/turns ratio; a step-up in voltage is always a step-down in current, and vice versa.
- An ideal transformer conserves power ($V_p I_p = V_s I_s$); real transformer losses (copper loss $I^2 R$, eddy current, hysteresis) are why efficiency $< 100\%$ — a common short-answer topic.

CHAPTER 8

Electromagnetic Waves

Formula Sheet + Tips & Tricks

Chapter 8: Electromagnetic Waves

Formula Sheet + Tips & Tricks

Formula Sheet

I. Displacement Current

Formula: $I_d = \varepsilon_0 A \frac{dE}{dt}$

I_d = displacement current, ε_0 = permittivity of free space, A = plate area, $\frac{dE}{dt}$ = rate of change of electric field.

II. Electromagnetic Waves

Formula (Speed in a medium): $v = \frac{c}{\sqrt{\mu_r \varepsilon_r}}$

v = speed of EM wave in medium, c = speed of light in vacuum, μ_r = relative permeability, ε_r = relative permittivity.

Formula (Wave equation): $c = f\lambda$

c = speed of light, f = frequency, λ = wavelength.

Formula (E_0 - B_0 relation): $B_0 = \frac{E_0}{c}$

E_0 = peak electric field, B_0 = peak magnetic field, c = speed of light.

Formula (Intensity): $I = \frac{1}{2} \varepsilon_0 c E_0^2$

I = intensity (average power per unit area), ε_0 = permittivity of free space, c = speed of light, E_0 = peak electric field.

Formula (Average energy density – electric): $\langle u_E \rangle = \frac{1}{4} \varepsilon_0 E_0^2$

$\langle u_E \rangle$ = time-averaged electric energy density; equal in magnitude to the average magnetic energy density.

Formula (Average energy density – magnetic): $\langle u_B \rangle = \frac{B_0^2}{4\mu_0} =$

$$\langle u_E \rangle$$

$\langle u_B \rangle$ = time-averaged magnetic energy density, μ_0 = permeability of free space.

Formula (Total average energy density): $\langle u \rangle = \langle u_E \rangle + \langle u_B \rangle = \frac{1}{2} \varepsilon_0 E_0^2$

$\langle u \rangle$ = total time-averaged energy density of the EM wave (electric + magnetic contributions, which are equal).

Tips, Tricks & Hacks

- Displacement current $I_d = \varepsilon_0 \frac{d\phi_E}{dt}$ was Maxwell's fix so that Ampere's law works even inside a charging capacitor's gap, where there's no conduction current but the field is still changing.
- Total current (conduction + displacement) is continuous around any loop — inside a capacitor gap, I_d exactly equals the conduction current in the wires, so charge is never "lost".
- In EM waves, $\vec{E}$, $\vec{B}$, and the direction of propagation are mutually perpendicular, and E , B oscillate *in phase* with each other (not 90° out of phase — a frequent misconception).
- Speed of EM wave in vacuum: $c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = \frac{E_0}{B_0}$ — this ratio E_0/B_0 equalling c is a favourite one-liner to test.
- Order of the EM spectrum by increasing frequency (decreasing wavelength): radio → microwave → infrared → visible → UV → X-ray → gamma ray — memorise this sequence, it's tested almost every year (uses/production/detection of each).
- All EM waves travel at speed c in vacuum regardless of frequency — what changes across the spectrum is wavelength/frequency and the method of production/detection, not the speed.
- Energy in an EM wave is carried equally by the electric and magnetic fields (average energy density of each is equal) — useful for intensity-related numericals.

CHAPTER 9

Ray Optics and Optical Instruments

Formula Sheet + Tips & Tricks

Chapter 9: Ray Optics and Optical Instruments

Formula Sheet + Tips & Tricks

Formula Sheet

I–II. Spherical Mirrors

Mirror Formula

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

u = object distance, v = image distance, f = focal length ($f = R/2$), all measured from pole.

Magnification (Mirror)

$$m = -\frac{v}{u} = \frac{f}{f - u}$$

Negative m = real, inverted; positive m = virtual, erect.

III. Snell's Law and Refractive Index

Snell's Law

$$n_1 \sin i = n_2 \sin r$$

i = angle of incidence, r = angle of refraction, n_1, n_2 = refractive indices of the two media.

Refractive Index

$$n = \frac{c}{v} = \frac{\sin i}{\sin r}$$

c = speed of light in vacuum, v = speed in medium.

IV. Apparent Depth

Apparent Depth

$$n = \frac{\text{Real Depth}}{\text{Apparent Depth}}$$

Valid for near-normal viewing from a rarer medium.

Normal Shift

$$\Delta t = t \left(1 - \frac{1}{n} \right)$$

t = thickness of slab/liquid; Δt = apparent upward shift of the object.

V. Total Internal Reflection**Critical Angle**

$$\sin C = \frac{1}{n}$$

n = refractive index of denser medium relative to the rarer medium; C = critical angle.

VI. Refraction at a Spherical Surface**Spherical Refraction**

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

n_1 = incident-side index, n_2 = refracted-side index, R = radius of curvature of the surface.

VII–VIII. Lens Maker's Formula**Lens Maker's Formula**

$$\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

n = refractive index of lens material relative to surrounding medium; R_1, R_2 = radii of the two surfaces.

Lens in a Liquid

$$\frac{1}{f_L} = \left(\frac{n_{\text{lens}}}{n_{\text{liquid}}} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

f_L = focal length of the lens when immersed in a liquid of index n_{liquid} .

IX–XI. Lens Formula, Magnification, Combination**Thin Lens Formula**

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

All distances measured from the optical centre.

Magnification (Lens)

$$m = \frac{v}{u} = \frac{f}{f + u}$$

Positive m = erect (virtual for convex, always virtual for concave); negative m = real, inverted.

Lenses in Contact

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$$

F = equivalent focal length of two thin lenses held in contact.

XII. Power of a Lens**Power**

$$P = \frac{1}{f \text{ (in metres)}}$$

Unit: dioptre (D). $P_{\text{total}} = P_1 + P_2$ for lenses in contact.

XIII. Combination of Lenses and Mirrors**Sequential Refraction/Reflection**

v_{final} found by applying lens/mirror formula stage by stage
Image of one element becomes the object for the next; effective mirror power $P_{\text{eff}} = P_{\text{lens}} + P_{\text{mirror}} + P_{\text{lens}}$ for lens-mirror silvered systems.

XIV–XVII. Prism**Prism Angle Relation**

$$A = r_1 + r_2$$

A = angle of prism, r_1, r_2 = angles of refraction at the two faces.

Deviation

$$\delta = i + e - A$$

i = angle of incidence, e = angle of emergence.

Refractive Index (Min. Deviation)

$$n = \frac{\sin \left(\frac{A + \delta_m}{2} \right)}{\sin \left(\frac{A}{2} \right)}$$

δ_m = angle of minimum deviation (occurs when $i = e$, $r_1 = r_2$).

Thin Prism Deviation

$$\delta = (n - 1)A$$

Valid for small prism angle A (thin prism approximation).

Angular Dispersion

$$\theta = (n_v - n_r)A$$

n_v, n_r = refractive indices for violet and red light.

XVIII–XIX. Microscopes**Simple Microscope**

$$M = 1 + \frac{D}{f} \text{ (image at } D\text{)}; \quad M = \frac{D}{f} \text{ (image at } \infty\text{)}$$

D = least distance of distinct vision (25 cm), f = focal length of lens.

Compound Microscope

$$M = m_o \times m_e = \frac{v_o}{u_o} \times \left(1 + \frac{D}{f_e} \right)$$

m_o = linear magnification of objective, m_e = angular magnification of eyepiece.

XX. Telescope**Astronomical Telescope**

$$M = -\frac{f_o}{f_e} \text{ (relaxed eye)}; \quad L = f_o + f_e$$

f_o, f_e = focal lengths of objective and eyepiece; L = tube length for image at infinity.

Telescope (Image at D)

$$M = -\frac{f_o}{f_e} \left(1 + \frac{f_e}{D} \right)$$

Used when the final image is formed at the least distance of distinct vision.

Tips, Tricks & Hacks

- Sign convention (Cartesian): distances measured against the incident light are negative, measured along it are positive; always fix the pole/optical centre as origin *before* substituting numbers — most mirror/lens errors come from skipping this.
- Mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ and lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ differ by a sign — write down which one you're using before you start, don't mix them mid-problem.
- For a concave mirror, remember the four zones by object distance vs f and $2f$: beyond C (real, inverted, diminished), at C (real, inverted, same size), between C and F (real, inverted, magnified), inside F (virtual, erect, magnified) — draw this number line once and you'll never forget it.
- Lens-maker's formula $\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$: if the lens is flipped or immersed in a denser medium, recompute n as the *relative* refractive index (lens w.r.t. surrounding medium), not the absolute one.
- Power of a lens $P = \frac{1}{f}$ (in metres) is additive for lenses in contact: $P = P_1 + P_2 + \dots$ — much faster than combining focal lengths directly, especially with more than 2 lenses.
- Critical angle and TIR: $\sin C = \frac{1}{n}$ — TIR only happens going from denser to rarer medium, and only when angle of incidence exceeds C ; this is the one-line justification for optical fibres, mirages, and sparkle in diamonds.
- Simple microscope magnification is $1 + \frac{D}{f}$ (image at near point) or $\frac{D}{f}$ (image at infinity) — the “+1” is the single most forgotten term in exams.
- Telescope magnifying power (normal adjustment) = $\frac{f_o}{f_e}$, and tube length = $f_o + f_e$ — for microscopes it's the objective's focal length that should be *small*, for telescopes it should be *large*; keep the two straight.

CHAPTER 10

Wave Optics

Formula Sheet + Tips & Tricks

Chapter 10: Wave Optics

Formula Sheet + Tips & Tricks

Formula Sheet

I. Reflection and Refraction of Light Waves

Speed in a medium

$$v = \frac{c}{n}$$

c = speed in vacuum, n = refractive index of medium, v = speed in medium

Time to traverse a thickness

$$t = \frac{d}{v}$$

d = thickness of medium, v = speed of light in that medium

Wavelength in a medium

$$\lambda_{\text{medium}} = \frac{\lambda_{\text{vacuum}}}{n}$$

Frequency does not change on entering a new medium; only v and λ change

Relative refractive index

$${}_A n_B = \frac{v_A}{v_B} = \frac{n_B}{n_A}$$

Refractive index of medium B with respect to medium A

Frequency–wavelength relation

$$f = \frac{c}{\lambda}$$

f = frequency, λ = wavelength in vacuum/air

II. Superposition of Waves

Intensity–amplitude relation

$$I \propto a^2$$

I = intensity, a = amplitude of the wave

Max. to min. intensity ratio

$$\frac{I_{max}}{I_{min}} = \left(\frac{a_1 + a_2}{a_1 - a_2} \right)^2$$

a_1, a_2 = amplitudes of the two superposing waves

Resultant intensity (phase form)

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$$

ϕ = phase difference between the two waves at the point of interest

III. Young's Double Slit Experiment (YDSE)**Fringe width**

$$\beta = \frac{\lambda D}{d}$$

D = slit-to-screen distance, d = slit separation, λ = wavelength

Position of n th bright fringe

$$x_n = \frac{n\lambda D}{d}$$

$n = 0, 1, 2, \dots$ measured from the central bright fringe

Position of n th dark fringe

$$x_n = \frac{(2n - 1)\lambda D}{2d}$$

$n = 1, 2, 3, \dots$ measured from the central bright fringe

Path difference

$$\Delta x = d \sin \theta \approx \frac{xd}{D}$$

θ = angle subtended at the slits; valid for small θ

Phase difference

$$\phi = \frac{2\pi}{\lambda} \Delta x$$

Relates path difference Δx to phase difference ϕ

IV. Angular Fringe Width

Angular position of n th bright fringe

$$\theta_n = \frac{n\lambda}{d}$$

Small-angle form of x_n/D

Angular fringe width

$$\Delta\theta = \frac{\lambda}{d}$$

Angular separation between two consecutive bright (or dark) fringes

V. Diffraction at a Single Slit

Minima condition

$$a \sin \theta = n\lambda$$

a = slit width, $n = 1, 2, 3, \dots$

Secondary (subsidiary) maxima

$$a \sin \theta = (2n + 1) \frac{\lambda}{2}$$

Approximate condition, $n = 1, 2, 3, \dots$

Angular width of central maximum

$$2\theta = \frac{2\lambda}{a}$$

Angle between the first minima on either side of the central maximum

Linear width of central maximum

$$w = \frac{2\lambda D}{a}$$

D = slit-to-screen distance (or focal length f of the focusing lens)

VI. Miscellaneous (Combined Relations)

Double-slit maxima inside single-slit envelope

$$N = \frac{2d}{e}$$

N = number of double-slit maxima within the central diffraction maximum, e = slit width, d = slit separation

Central maximum width using a lens

$$w = \frac{2f\lambda}{a}$$

Screen placed at focal plane of a convex lens of focal length f kept close to the slit

Tips, Tricks & Hacks

- Constructive interference: path difference = $n\lambda$; destructive: path difference = $(n + \frac{1}{2})\lambda$ — always compute path difference first, then just check which family it belongs to.
- Fringe width $\beta = \frac{\lambda D}{d}$: it increases with wavelength and distance to screen, decreases as slit separation increases — use this to instantly judge how fringe pattern changes if any one variable changes, without recomputing.
- If the whole YDSE apparatus is immersed in a medium of refractive index n , wavelength shrinks to λ/n , so fringe width shrinks to β/n — a very commonly tested modification.
- Diffraction from a single slit: central maximum is *twice* as wide as the secondary maxima, and its width = $\frac{2\lambda D}{a}$ — don't confuse this with the double-slit fringe width formula, they look similar but differ by that factor of 2 and by d vs a .
- Condition for minima in single-slit diffraction is $a \sin \theta = n\lambda$ (note: this is a minima condition here, unlike YDSE where $n\lambda$ gave maxima) — this sign/role flip is the single most common mix-up between the two topics.
- Resolving power of an optical instrument improves with *shorter* wavelength and *larger* aperture — this is why UV/electron microscopes resolve finer detail than visible-light ones.
- Polarisation only happens to transverse waves; sound (longitudinal) can never be polarised — a quick way to settle “can X be polarised” conceptual questions.
- Malus's law $I = I_0 \cos^2 \theta$: intensity through a second polariser depends on $\cos^2$ of the angle between the two polariser axes — remember it's $\cos^2$, not $\cos$, a very common careless error.

CHAPTER 11

Dual Nature of Radiation and Matter

Formula Sheet + Tips & Tricks

Chapter 11: Dual Nature of Radiation and Matter

Formula Sheet + Tips & Tricks

Formula Sheet

Constants used in this chapter:

$$h = 6.6 \times 10^{-34} \text{ J s} \quad c = 3 \times 10^8 \text{ m/s} \quad e = 1.6 \times 10^{-19} \text{ C}$$

$$m_e = 9.1 \times 10^{-31} \text{ kg} \quad m_p = m_n = 1.67 \times 10^{-27} \text{ kg} \quad 1 \text{ Å} = 10^{-10} \text{ m}$$

I. Photon Energy, Momentum, Work Function

Photon Energy

$$E = h\nu = \frac{hc}{\lambda}$$

ν = frequency, λ = wavelength

Photon Momentum

$$p = \frac{E}{c} = \frac{h}{\lambda}$$

momentum carried by a photon

Dynamic Mass of Photon

$$m = \frac{E}{c^2}$$

rest mass of a photon is always zero

Threshold Frequency

$$\nu_0 = \frac{W_0}{h}$$

W_0 = work function of the metal

Threshold Wavelength

$$\lambda_0 = \frac{hc}{W_0}$$

longest wavelength that can cause emission

Photon Count from Power

$$n = \frac{P}{E_{\text{photon}}}$$

P = power of source, n = photons/s

II. Einstein's Photoelectric Equation**Einstein's Equation**

$$E_{k(\text{max})} = h\nu - W_0 = \frac{hc}{\lambda} - W_0$$

$E_{k(\text{max})}$ = max. KE of ejected electron

Max. Speed of Photoelectron

$$v_{\text{max}} = \sqrt{\frac{2E_{k(\text{max})}}{m_e}}$$

from max. kinetic energy

III. Stopping Potential**Stopping Potential Relation**

$$eV_0 = h\nu - W_0$$

V_0 = stopping potential (volt)

Slope of V_0 - ν Graph

$$\text{slope} = \frac{h}{e}$$

gives Planck's constant from graph

Intercepts of V_0 - ν Graph

$$\nu_0 = \frac{W_0}{h}, \quad V_0|_{\nu=0} = -\frac{W_0}{e}$$

ν_0 = x-intercept, $-W_0/e$ = y-intercept

IV, V & VI. de Broglie Wavelength**de Broglie Wavelength (general)**

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

p = momentum of the particle

de Broglie Wavelength (KE given)

$$\lambda = \frac{h}{\sqrt{2mE_k}}$$

E_k = kinetic energy of the particle

de Broglie Wavelength (electron, accelerating p.d. V)

$$\lambda = \frac{h}{\sqrt{2m_e eV}} \approx \frac{12.27}{\sqrt{V}} \text{ \AA}$$

V in volts, standard result for electrons

Tips, Tricks & Hacks

- Work function $\phi_0 = h\nu_0$ sets the threshold frequency; below ν_0 , no photoelectrons are emitted *no matter how intense* the light — intensity controls number of photoelectrons, frequency controls their maximum kinetic energy, never mix the two roles up.
- Einstein's photoelectric equation: $KE_{max} = h\nu - \phi_0 = eV_0$ — plotting V_0 vs ν gives a straight line of slope h/e and intercept related to ϕ_0 ; this graph is a favourite numerical/derivation.
- Stopping potential depends only on frequency of incident light, not intensity — increasing intensity increases photocurrent (saturation current) but leaves V_0 unchanged at fixed frequency.
- de Broglie wavelength $\lambda = \frac{h}{p}$ applies to *all* matter, not just electrons — for a particle accelerated through potential V , use $\lambda = \frac{h}{\sqrt{2mqV}}$ directly instead of finding v first, it saves a step.
- For electrons specifically, the handy shortcut is $\lambda = \frac{1.227}{\sqrt{V}} \text{ nm}$ (V in volts) — worth memorising to skip algebra in MCQs.
- Heavier particles (protons, alpha particles) have much shorter de Broglie wavelength than electrons for the same kinetic energy or same accelerating voltage, since $\lambda \propto \frac{1}{\sqrt{m}}$ — a quick comparison trick for ranking questions.
- Davisson–Germer experiment is the experimental proof of de Broglie's hypothesis (electron diffraction) — commonly asked as a one-line “which experiment confirms...” question.

CHAPTER 12

Atoms

Formula Sheet + Tips & Tricks

Chapter 12: Atoms

Formula Sheet + Tips & Tricks

Formula Sheet

I. α -Scattering Experiments

$$r_0 = \frac{1}{4\pi\epsilon_0} \frac{Z_1 Z_2 e^2}{E}$$

r_0 = distance of closest approach, Z_1, Z_2 = charge numbers, E = kinetic energy of incident particle

II. Radii of Orbits and Velocities

$$r_n = n^2 r_1, \quad r_1 = 0.529 \text{ \AA (for hydrogen, } Z = 1)$$

r_n = radius of n th Bohr orbit; radius scales as n^2/Z and inversely with orbiting particle's mass

II. Radii of Orbits and Velocities

$$v_n = \frac{e^2}{2\epsilon_0 n h}$$

v_n = orbital speed in n th orbit; independent of the orbiting particle's mass

II. Radii of Orbits and Velocities

$$T = \frac{2\pi r_n}{v_n}$$

T = time period of revolution in the n th orbit

III. Energy of Electron in Different Orbits

$$E_n = -\frac{13.6 Z^2}{n^2} \text{ eV}$$

E_n = total energy in n th orbit; for a Bohr orbit, $KE = -E_n$ and $PE = 2E_n$

IV/V. Transitions, Excitation and Ionisation

$$\Delta E = E_i - E_f = h\nu = \frac{hc}{\lambda}$$

ΔE = photon energy absorbed/emitted; ν = frequency, λ = wavelength

V. Excitation and Ionisation Potentials

$$E_{\text{ionisation}} = -E_1 = 13.6 Z^2 \text{ eV}$$

Ionisation energy = energy to take electron from ground state ($n = 1$) to $n = \infty$; numerically equal to ionisation potential in volts

VI. Series in Hydrogen Spectrum

$$\frac{1}{\lambda} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

R = Rydberg constant; n_1 = lower level (series), $n_2 > n_1$ = upper level

VI. Series in Hydrogen Spectrum

$$E_{\text{ionisation}} = Rhc$$

Series-limit (ionisation) energy obtained by putting $n_1 = 1$, $n_2 \rightarrow \infty$

Tips, Tricks & Hacks

- Rutherford's α -scattering: most particles pass straight through (atom is mostly empty space), a few deflect at large angles (dense, positively charged nucleus) — know which observation supports which conclusion, it's asked as a match-the-following.
- Bohr radius scales as $r_n \propto \frac{n^2}{Z}$ and orbital speed as $v_n \propto \frac{Z}{n}$ — as n increases, orbits get bigger and slower; as Z increases (heavier hydrogen-like ion), orbits shrink and speed up.
- Energy of electron in n th orbit: $E_n = \frac{-13.6 Z^2}{n^2} \text{ eV}$ — memorise 13.6 eV for hydrogen ground state, then just scale by Z^2/n^2 for any hydrogen-like ion or excited state.
- Energy released/absorbed on transition $n_2 \rightarrow n_1$ uses $\Delta E = 13.6 Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{ eV}$ — for emission $n_2 > n_1$ (energy released, photon emitted); for absorption $n_1 < n_2$ needs energy input.
- Ionisation energy is the energy to take an electron from its current level to $n = \infty$ (i.e., $E = 0$); excitation energy is between two *bound* levels — don't confuse the two when a question says "minimum energy to ionise" vs "minimum energy to excite to next level".
- Spectral series memory: Lyman (UV, ends at $n = 1$), Balmer (visible, ends at $n = 2$), Paschen/Brackett/Pfund (IR, end at $n = 3, 4, 5$) — "Lyman is Lowest ($n = 1$), Balmer is visible (the one you can actually see)" is a quick recall hook.
- Maximum number of spectral lines from an electron de-exciting from level n is

$\frac{n(n-1)}{2}$ — a very frequently tested combinatorics-style formula.

CHAPTER 13

Nuclei

Formula Sheet + Tips & Tricks

Chapter 13: Nuclei

Formula Sheet + Tips & Tricks

Formula Sheet

Standard Constants used in this Chapter:

$$m_p = 1.00728 \text{ u} \quad m_n = 1.00867 \text{ u} \quad m_e = 0.00055 \text{ u}$$

$$h = 6.6 \times 10^{-34} \text{ J s} \quad c = 3 \times 10^8 \text{ m s}^{-1}$$

$$1 \text{ MeV} = 1.60 \times 10^{-13} \text{ J} \quad 1 \text{ u} = 931.5 \text{ MeV (i.e. } 1 \text{ u} \times c^2 = 931.5 \text{ MeV)}$$

I. Mass-Energy Relation

Mass-Energy Equivalence

$$E = mc^2$$

E = energy released/absorbed (J); m = mass converted (kg); c = speed of light

Percentage Change in Mass

$$\% \text{ increase} = \frac{\Delta m}{M} \times 100$$

Δm = mass gained/lost; M = original mass of body

II. Pair-Production and Annihilation of Matter

Minimum Energy for Pair Production

$$E_{min} = 2m_e c^2$$

m_e = rest mass of electron (equal to rest mass of positron)

Photon Energy-Wavelength Relation

$$E = \frac{hc}{\lambda}$$

h = Planck's constant; λ = wavelength of the photon

Particle-Antiparticle Annihilation Energy

$$E = 2mc^2$$

m = rest mass of each particle (e.g. proton and antiproton)

III. Mass Defect and Binding Energy

Mass Defect

$$\Delta m = [Zm_p + (A - Z)m_n] - M_{nucleus}$$

Z = atomic number; A = mass number; $M_{nucleus}$ = actual nuclear mass

Binding Energy

$$BE = \Delta m \text{ (in u)} \times 931.5 \text{ MeV}$$

Δm expressed in atomic mass units (u)

Binding Energy per Nucleon

$$\frac{BE}{A}$$

Measure of nuclear stability; higher value $\Rightarrow$ more stable nucleus

IV. Q-Value of Nuclear Reactions

Q-Value

$$Q = (\Sigma m_{reactants} - \Sigma m_{products}) \times 931.5 \text{ MeV}$$

Positive Q = energy released (exothermic); Negative Q = energy absorbed

V. Nuclear Fission

Fission Energy

$$E = \Delta mc^2$$

Δm = mass defect in the fission process

Energy-kWh Conversion

$$1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$$

Used to express large nuclear energies in electrical units

Fission Rate

$$\frac{dN}{dt} = \frac{P}{E_{fission}}$$

P = power output (W); $E_{fission}$ = energy released per fission (J)

VI. Nuclear Fusion

Fusion Energy

$$E = \Delta mc^2$$

Δm = mass defect between fusing nuclei and product nucleus

Useful Energy with Efficiency

$$E_{\text{useful}} = \eta \times E_{\text{total}}$$

η = fractional efficiency of the generator/reactor

Tips, Tricks & Hacks

- Mass defect $\Delta m = [Zm_p + (A - Z)m_n] - M_{\text{nucleus}}$ is always positive for a stable nucleus — this “missing mass” converted to energy is precisely what binds the nucleus together.
- Binding energy per nucleon peaks around mass number $A \approx 56$ (iron region) — nuclei lighter than this release energy on *fusion* (moving right, up the curve), nuclei heavier release energy on *fission* (splitting to move up the curve too); both processes move nuclei toward the peak.
- $1 \text{ u} = 931.5 \text{ MeV}/c^2$ — memorise this conversion factor, it’s used in virtually every mass-defect/Q-value numerical.
- Q-value of a reaction = (mass of reactants – mass of products) c^2 — positive Q means energy is released (exothermic, reaction proceeds spontaneously), negative Q means energy must be supplied.
- Radioactive decay is a random, spontaneous process independent of external conditions (pressure, temperature) — a common conceptual trap is expecting decay rate to change with the environment; it doesn’t.
- Half-life and mean life are related by $T_{1/2} = 0.693 \tau$ — mean life is always *longer* than half-life; don’t accidentally swap them in a numerical.
- Activity $A = \lambda N = A_0 e^{-\lambda t}$ decays with the *same* exponential law as N itself — so activity halves every half-life too, just like the number of undecayed nuclei.
- Pair production needs a photon of energy $\geq 2m_e c^2 = 1.02 \text{ MeV}$ (creates an electron-positron pair); annihilation is the reverse, producing (usually two) photons of 0.51 MeV each — a nice paired fact to remember together.

CHAPTER 14

Semiconductor Electronics

Formula Sheet + Tips & Tricks

Chapter 14: Semiconductor Electronics

Formula Sheet + Tips & Tricks

Formula Sheet

I. p-n Junction Diode – Forward Bias

Formula: $V_{\text{battery}} = I_{\text{max}}R + V_D$

V_{battery} = battery EMF, I_{max} = maximum safe forward current, R = series resistance, V_D = constant forward voltage drop across the diode.

Formula: $R_{\text{dynamic}} = \frac{\Delta V}{\Delta I}$

R_{dynamic} = dynamic (AC) resistance of the diode, ΔV = change in voltage across diode, ΔI = corresponding change in diode current.

Formula: $I = \frac{V - V_0}{R}$

I = current in the circuit, V = applied battery voltage, V_0 = barrier (junction) potential of the diode, R = series resistance.

Tips, Tricks & Hacks

- In forward bias, the depletion layer narrows and resistance drops (diode conducts easily); in reverse bias, it widens and resistance rises (diode blocks, except for a tiny leakage current) — state cause (depletion width change) before effect (current) for full marks.
- Diode symbol arrow points in the direction of *conventional current flow* when forward biased (p to n externally, i.e. from p-side terminal, through diode, to n-side) — a quick way to read circuit diagrams instantly.
- Zener diode is operated deliberately in *reverse breakdown* (unlike a normal diode, which you avoid pushing into breakdown) — this is precisely what makes it useful as a voltage regulator.
- A full-wave rectifier (using two diodes or a bridge of four) gives twice the output frequency and higher average output than a half-wave rectifier — know this comparison, it's a classic short-answer/diagram question.
- For a p-n junction, the barrier potential is typically $\sim 0.7\text{ V}$ for silicon and $\sim 0.3\text{ V}$ for germanium — if a question doesn't specify, silicon values are the safer default

assumption in Indian board exams.

- Photodiode is operated in reverse bias (so that a small change in reverse current due to light can be measured clearly against a low, stable background current) — remembering “reverse bias for sensing, forward bias for conducting” separates diode-as-sensor from diode-as-rectifier questions.